Total No. of Questions : 8]

SEAT No. :

PC-2446

[Total No. of Pages : 2

[6354]-574

B.E. (Information Technology)

DEEP LEARNING

(2019 Pattern) (Semester - VII) (414443)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) What is Recurrent Neural Network (RNN)? Draw and Explain Simple Recurrent Neural Network. Explain four types of RNN with proper diagram. **[8]**

b) Explain in detail the following advantages of RNN over other types of Neural Network. **[9]**

i) Ability To Handle Variable-Length Sequences

ii) Memory of Past Inputs

OR

Q2) a) Discuss how the RNNs suffer from the problem of vanishing gradients. And how is it overcome with Long Short-Term Memory Network (LSTM)? **[9]**

b) What is meant by Recursive Neural Network? How is it different from Recurrent Neural Network? **[8]**

Q3) a) What is Autoencoder? Elaborate its two networks Encoder and Decoder. **[5]**

b) How do autoencoders differ from other unsupervised learning techniques like PCA or clustering algorithms? **[5]**

c) Explain the following listed hyperparameters and methods which are used to optimize the training of autoencoders : **[8]**

i) Activation functions ii) Loss functions iii) Learning rate iv) Batch size v) Early stopping vi) Initialization

P.T.O.

OR

Q4) a) Explain the working of denoising autoencoder. In what way, it differ from regular autoencoders? **[5]**

b) What is a contractive autoencoder? How does it differs from regular autoencoders? List the most appropriate applications of contractive autoencoders. **[8]**

c) Differentiate between Undercomplete Autoencoder and Sparse Autoencoder. **[5]**

Q5) a) What is a DenseNet? Why do we need Densenet? List it's advantages. **[5]**

b) What is transfer learning in CNN? **[7]**

c) Categorize Domain Adaptation methods - Homogeneous DA, Heterogeneous DA. **[5]**

OR

Q6) a) What are the representation learning challenges? **[5]**

b) "Is Densenet good for image classification?" Extend your idea about it. **[5]**

c) What is the general framework of Transfer Learning? **[7]**

Q7) a) Explain content based, collaborative and hybrid recommender system with pros and cons. **[9]**

b) State different applications of Deep Learning? With suitable diagram explain use of CNN for image classification. **[9]**

OR

Q8) a) State applications of NLP. Why RNN is preferred as compare to CNN for Natural Language Processing? State different NLP tasks where RNN is used. **[9]**

b) State and Explain different centrality measures used in social network analysis. **[9]**